API 676 3rd Edition Comparative Summary – Section 6 – Rev. 14

| *2nd Edition* | *3rd Edition* |
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| **SECTION 2—BASIC DESIGN**<br>**2.1 General** | **6 Basic Design**<br>**6.1 General** |
| 2.1.1 The equipment (including auxiliaries) covered by this standard shall be designed and constructed for minimum service life of 20 years and at least three years of uninterrupted operation. It is recognized that this is a design criterion. | 6.1.1 The equipment (including auxiliaries, but excluding normal maintenance shall be designed and constructed for a minimum service life of 20 years and at least 3 years of uninterrupted operation. It is recognized that these requirements are design criteria, and that service or duty severity, mis-operation or improper maintenance can result in a machine failing to meet these criteria.<br>The term 'design' shall apply to parameters or features of the equipment supplied by the manufacturer. The term 'design' should not be used in the purchaser's enquiry or specification because it can cause confusion in understanding the order. |
| 2.1.2 The vendor shall assume unit responsibility for all equipment, including pump, driver, power transmission, and all auxiliary systems included in the scope of the order. | |
| 2.1.3 The purchaser will specify the equipment's normal operating point on the data sheets. If a range of operating conditions is specified, the pump vendor shall advise the purchaser about the pump's minimum and maximum capacity at its rated differential pressure and its required brake horsepower. Anticipated process variations that may affect the sizing of the pump and the driver (such as changes in pressure, temperature, or properties of fluids handled, and special plant startup conditions) will be specified. | 6.1.2 • The purchaser shall specify the normal operating point and all other required operating points. If a range of operating conditions is specified, the pump supplier shall advise the purchaser about the pump's minimum and maximum capacity at its rated differential pressure and its required brake horsepower. Anticipated process variations that may affect the sizing of the pump and the driver (such as changes in pressure, temperature, or properties of fluids handled, and special plant startup conditions) shall be specified.<br><br>6.1.3 ● Pumps shall be designed for flammable or hazardous services. If specified that the pump is to be used in non-flammable or non-hazardous services, the supplier may provide an alternate design. |
| 2.1.4 Control of the sound pressure level (SPL) of all equipment furnished shall be a joint effort of the purchaser and the vendor. The equipment furnished by the vendor shall conform to the maximum allowable sound pressure level specified by the purchaser. | 6.1.4 Control of the sound pressure level (SPL) of all equipment supplied shall be a joint effort of the purchaser and the supplier having unit responsibility. The equipment provided by the supplier shall conform to the maximum allowable sound pressure level specified. In order to determine compliance, the supplier shall provide both maximum sound pressure and sound power level data per octave band for the equipment. |

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| 2.1.5 Equipment shall be designed to run without damage to the trip speed and relief valve settings. | 6.1.5 Equipment shall be selected to run simultaneously at the pressure-limiting accumulation pressure and at trip speed without suffering damage. Supplier shall advise purchaser of increased power operating requirements necessary to achieve this.<br><br>6.1.6 For direct driven equipment the equipment's maximum continuous operating speed shall be not less than 105% of the rated speed for adjustable speed machines and shall be equal to the rated speed for constant speed motor drives.<br><br>6.1.7 For gear-driven equipment the gearbox input shaft maximum continuous operating speed shall not be less than 105% of the rated speed for adjustable speed machines and shall be equal to the rated speed for constant speed motor drives.<br><br>6.1.8 The equipment's trip speed shall not be less than the following percentages of maximum continuous speed:<br>Adjustable speed motor 110%<br>Constant speed motor 100%<br>Reciprocating engine 110% |
| 2.1.6 The arrangement of the equipment, including piping and auxiliaries, shall be developed jointly by the purchaser and the vendor. The arrangement shall provide adequate clearance areas and safe access for operation and maintenance. | 6.1.9 The arrangement of the equipment, including piping and auxiliaries, shall be developed jointly by the purchaser and the supplier. The arrangement shall provide adequate clearance areas and safe access for operation and maintenance. |
| 2.1.7 Motors, electrical components, and installations shall be suitable for the area classification (class, group, and division or zone) specified by the purchaser on the data sheets and shall meet the requirements of NFPA 70, Articles 500, 501, 502 and 504, as well as local codes specified and furnished by the purchaser. | • 6.1.10 Motors, electrical components, and electrical installations shall be suitable for the area classification (class, group, and division or zone) specified by the purchaser and shall meet the requirements of the applicable sections of IEC 60079, or NFPA 70, API RP 500, and NEC as specified, as well as any local codes specified and supplied by the purchaser. |
| 2.1.8 Oil reservoirs and housings that enclose moving lubricated parts (such as bearings, shaft seals, highly polished parts, instruments, and control elements) shall be designed to minimize contamination by moisture, dust, and other foreign matter during periods of operation and idleness. | 6.1.11 Oil reservoirs and housings that enclose moving lubricated parts such as bearings, shaft seals, highly polished parts, instruments, and control elements shall be designed to minimize contamination by moisture, dust, and other foreign matter during periods of operation and idleness. |
| 2.1.9 All equipment shall be designed to permit rapid and economical maintenance, particularly regarding packing and seals. Major parts shall be designed (shouldered or | 6.1.12 All equipment shall be designed to permit rapid and economical maintenance, particularly regarding packing and seals. Major parts shall such as casing components and |

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| cylindrically doweled) and manufactured to ensure accurate alignment on reassembly. | bearing housings shall be designed and manufactured to ensure accurate alignment on reassembly. This may be accomplished by the use of shouldering, cylindrical dowels, or keys. |
| 2.1.10 The machine and its driver shall perform on the test stand and on their permanent foundation within the specified acceptance criteria. After installation, the combined performance of the units shall be the joint responsibility of the purchaser and the vendor having unit responsibility. | 6.1.13 The equipment (machine, driver, and auxiliary equipment) shall perform on the test stand and on its permanent foundation within the specified test tolerances (see 8.3.6). After installation, the performance of the combined units shall be the joint responsibility of the purchaser and the supplier who has unit responsibility.<br>Note: Many factors can adversely affect performance of the pump at site. These factors include piping layout, piping connection loads, alignment at operating conditions, support structure, handling during shipment and handling and assembly on site. |
| 2.1.11 The purchaser will specify whether the installation is indoors (heated or unheated) or outdoors (with or without a roof), as well as the weather and environmental conditions in which the equipment must operate (including maximum and minimum temperatures, unusual humidity, and dusty or corrosive conditions). | • 6.1.14 The equipment, including all auxiliaries, shall be suitable for operation under the environmental conditions specified by the purchaser. The environmental conditions shall include whether the installation is indoors (heated or unheated) or outdoors (with or without a roof), maximum and minimum temperatures, unusual humidity, dusty, or corrosive conditions.<br><br>6.1.15 The equipment, including all auxiliaries shall be suitable for operation using the utility conditions specified. |
| 2.1.12 Spare parts for the machine and all furnished auxiliaries shall meet all the criteria of this standard. | 6.1.16 Spare and replacement parts for the machine and all furnished auxiliaries shall meet all the criteria of this standard. |
| 2.2.4 Bolting shall be furnished as specified in 2.2.4.1 through 2.2.4.5. | 6.1.17 Bolting shall conform to 6.1.17.a) through 6.1.17.d). |
| 2.2.4.1 The details of threading shall conform to ASME B1.1. | a) details of threading shall conform to ISO 261, ISO 262, ISO 724, and ISO 965 or to ASME B1.1 |
| 2.2.4.2 Studs shall be supplied unless cap screws are specifically approved by the purchaser. | b) unless specifically approved by the purchaser, internal socket type, slotted nut or spanner type bolting shall not be used |
| 2.2.4.3 Adequate clearance shall be provided at bolting locations to permit the use of socket or box wrenches. | c) fasteners (excluding washers and headless set-screws) shall have the material grade and manufacturer's identification symbols applied to one end of studs 10 mm (3/8 in) in diameter and larger and to the heads of bolts 6 mm (1/4 in) in diameter and larger. If the available area is inadequate, the grade symbol may be marked on one end and the |
| 2.2.4.4 Internal socket-type, slotted-nut, or spanner-type bolting shall not be used unless specifically approved by the purchaser. | |

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| 2.2.4.5 Stud ASTM grade marking shall be located on the nut end of the exposed stud end. | manufacturer's identification symbol marked on the other end. Studs shall be marked on the exposed end.<br><br>d) Metric fine and UNF threads shall not be used.<br><br>Note: A set-screw is a headless screw with an internal hexagonal opening in one end.<br><br>6.1.18 The pump mounting surfaces shall meet the following criteria:<br>a) They shall be machined to a finish of 6.3 µm (250 micro-inches) arithmetic average roughness (Ra) or smoother.<br>b) To prevent a soft foot, they shall be in the same horizontal plane within 25 µm (0.001 in.).<br>c) Each mounting surface shall be machined within a flatness of 1:2000 corresponding surfaces shall be in the same plane within 150 µm/m (0.002 in/ft).<br>d) The upper machined or spot faced surface shall be parallel to the mounting surface.<br>e) Hold-down bolt holes shall be drilled perpendicular to the mounting surface or surfaces, machined or spot faced to a diameter three times that of the hole and to allow for equipment alignment. For hold down bolts 1" and larger the hole shall be 15 mm (1/2") larger in diameter than the hold down bolt. For bolts less than 1" in diameter, the holes shall be ¼" larger in diameter than the bolt. |
| | **6.2 Selection and Rating of Pump Type** |
| 2.1.13 Unless otherwise specified, the vendor shall recommend the pump speed for the specified service, considering such factors as NPSH, maximum fluid viscosity, solids and abrasives content, and wear allowance if required.<br><br>Note: It must be recognized that different rotary pump designs operate on different principles so that no one speed criterion can be applied. | 6.2.1 Unless otherwise specified, the supplier shall recommend the pump speed for the specified service, considering such factors as NPSHA, NPSHR, maximum fluid viscosity, solids and abrasives content, and wear allowance if required.<br><br>Note: It must be recognized that different rotary pump designs operate on different principles so that no one speed criterion can be applied. |
| **2.2 Pressure Casings** | **6.3 Pressure-Containing and Pressure-Retaining Parts** |
| 2.2.1 The hoop-stress values used in the design of the casing shall not exceed the maximum allowable stress values in tension specified in Section VIII, Division 1, of the ASME Code at the maximum operating temperature of the material used. | 6.3.1 The pressure-containing parts shall be designed in accordance with 6.3.1.1 (or 6.3.1.2, as selected by the supplier) and 6.3.1.3 to achieve the following:<br><br>a) Operate without leakage while subject simultaneously to the MACP (and corresponding |

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| 2.2.2 The maximum allowable working pressure of the casing shall be at least equal to the specified relief valve setting. The relief valve setting shall exceed the rated discharge pressure by at least 10 percent or 1.7 bar (25 psi), whichever is greater. | temperature) and the worst case combination of maximum allowable nozzle loads applied to all nozzles.<br><br>b) Withstand the hydrostatic test in paragraph 8.3.2.<br><br>6.3.1.1 Pressure-containing components may be designed with the aid of finite element analysis provided that the value of the stress intensity reflects a requirement to perform a hydro test at 150% of MACP.<br><br>6.3.1.2 The allowable tensile stress used in the design of the pressure components, excluding bolting, for any material shall not exceed 0.25 times the minimum ultimate tensile strength or 0.67 times the minimum yield strength for the material whichever is lower across the full range of specified operating temperature.<br><br>6.3.1.3 For casing joint bolting, the allowable stress, as determined in paragraph 6.3.1.2 shall be used to determine the total bolting area based on hydrostatic load and gasket preload, as applicable. The preload stress shall not exceed 0.75 times the bolting material minimum yield strength.<br><br>Note: Preloading the bolting is required to prevent unloading the bolted joint due to cyclic operation.<br><br>6.3.1.4 A corrosion allowance of at least 3 mm (0.12 in) shall be added to the casing thickness used in 6.3.1. The corrosion allowance shall also be added to all auxiliary connections exposed to the same fluid as the pressure-containing casing.<br><br>Note: The supplier may propose alternative corrosion allowances for consideration if materials of construction with superior corrosion resistance are employed.<br><br>6.3.2 • The purchaser shall install a pressure-limiting valve on each positive displacement pump. If specified, this valve shall be provided by the supplier. The pressure-limiting valve accumulation pressure shall not exceed the maximum allowable working pressure of the pump.<br><br>6.3.3 Casing and other pressure-retaining parts and supports shall be designed to prevent detrimental distortion caused by the worst combination of temperature, pressure, torque, |

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| | and allowable external forces and moments based on the specified operating conditions.<br><br>6.3.3.1 Unless otherwise specified, the suction region and the discharge region of the pump shall be designed for different pressure ratings.<br><br>6.3.3.2 • If specified, suction regions shall be designed for the same MACP as the discharge section.<br><br>6.3.3.3 Casings shall be sealed using flat gaskets; confined, controlled compression spiral wound gaskets; or O-rings. |
| 2.2.3 The use of tapped holes in pressure parts shall be minimized To prevent leakage in pressure sections of casings, metal equal in thickness to at least half of the nominal bolt diameter, in addition to the allowance for corrosion, shall be left around and below the bottom of drilled and tapped holes. The depth of the tapped holes shall be at least 1 1⁄2 times the stud diameter. | 6.3.4 The use of threaded holes in pressure-retaining parts shall be minimized. To prevent leakage in these parts, metal equal in thickness to at least half the nominal bolt diameter (including the allowance for corrosion) shall be left around and below the bottom of drilled and threaded holes. The depth of the threaded holes shall be at least 1.5 times the stud diameter. |
| 2.2.5 Jackscrews, lifting lugs, eyebolts, guide dowels, and alignment dowels shall be provided to facilitate disassembly and reassembly when required by pump design. When jackscrews are used as a means of parting contacting faces, one of the faces shall be relieved (counter bored or recessed) to prevent a leaking joint or improper fit caused by marring. Guide rods shall be of sufficient length to prevent damage to the internals or studs by any component during disassembly and reassembly. Lifting lugs or eyebolts shall be provided for lifting only the top half of the casing. | 6.3.5 If required by the pump design, jackscrews, cylindrical alignment dowels and/or other appropriate devices shall be provided to facilitate disassembly. If jackscrews are used as a means of parting contacting faces, one of the faces shall be relieved (counter-bored or recessed) to prevent a leaking joint or an improper fit caused by marring of the face. |
| 2.2.6 Pump cooling or heating jackets, if required, shall be designed to positively prevent process fluid from leaking into the coolant. When cooling of casings is necessary, separate, non interconnecting chambers are required for casing and head. | 6.3.6 If jacketed pump casings are required, they shall be designed to positively prevent process fluid from leaking into the jacket. |
| 2.2.7 Unless otherwise specified, jackets shall be designed for minimum of 5.2 bar gauge (75 psig) working pressure and shall be suitable for hydrostatic testing at a minimum of 8 bar gauge (115 psig). 2.2.8 The equipment feet shall be provided with vertical jackscrews and shall be drilled with pilot holes that are accessible for use in final doweling. | |
| **2.3 Casing Connections** | **6.4 Casing Connections** |

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| | 6.4.1 Provision shall be made for complete venting (unless it is self venting) and draining of the pump and systems provided by the supplier.<br><br>6.4.2 All pumps shall be provided with vent and drain connections, except that vent connections may be omitted if the pump is made self-venting by the arrangement of the nozzles.<br><br>6.4.2.1 If the pump cannot be completely drained for geometrical reasons, this shall be stated in the proposal.<br><br>6.4.2.2 The operating manual shall include a drawing indicating the quantity and location(s) of the fluid remaining in the pump.<br><br>Note: A pump is considered functionally self-venting if the nozzle arrangement and the casing configuration permit sufficient venting of gases to prevent loss of prime during the starting sequence.<br><br>6.4.3 Casing opening sizes |
| 2.3.1 Inlet and outlet connections shall be flanged or machined and studded, and oriented as specified. If threaded connections are standard, they may be used for NPS 1⁄2 and smaller. The inlet connection shall be rated for the maximum allowable inlet working pressure. The outlet connection shall be rated for the maximum allowable outlet working pressure. (2.2.2) The purchaser will specify if higher inlet connection flange ratings are required or if threaded connections NPS 1⁄2 and smaller are prohibited. | 6.4.3.1 Openings for nozzles and other pressure casing connections shall be standard pipe sizes in accordance with ISO 6708 or ANSI B16.5. Openings of DN 32, 65, 90, 125, 175 and 225 (NPS 1-1/4, 2-1/2, 3-1/2, 5, 7 and 9) shall not be used. |
| 2.3.3 Casing openings for piping connections shall be at least NPS 1⁄2 and shall be flanged or machined and studded. Where flanged or machined and studded openings are impractical, threaded openings in sizes NPS 1⁄2 through NPS 1 1⁄2 are permissible. These threaded openings shall be installed as specified in 2.3.3.1 through 2.3.3.7. | 6.4.3.2 Casing connections other than suction and discharge nozzles shall be at least DN 15 (NPS 1/2) for pumps with discharge nozzle openings DN 50 (NPS 2) and smaller. Connections shall be at least DN 20 (NPS 3/4) for pumps with discharge nozzle openings DN 80 (NPS 3) and larger, except that connections for seal flush piping and gauges may be DN 15 (NPS 1/2) regardless of pump size.<br><br>6.4.4 Suction and discharge nozzles |
| | 6.4.4.1 Suction and discharge nozzles shall be flanged or machined and studded for sizes DN 50 (NPS 2) and larger. Sizes DN 40 (NPS 1-1/2) and smaller may be threaded |

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| 2.3.2 Connections welded to the casing shall meet the material requirements of the casing, including impact values, rather than the requirements of the connected piping (2.9.4.5). | connections.<br><br>Note: When machined and studded suction and discharge nozzles are provided, purchaser shall supply and install piping spool pieces near the pump, so that large sections of piping do not have to be disassembled in order to remove the pump for a major overhaul.<br><br>6.4.4.2 • If specified, all connections shall be suitable for the MACP.<br><br>**6.5 Auxiliary Connections**<br><br>6.5.1 Connections welded to the casing shall meet the material requirements of the casing including impact values, rather than the requirements of the connected piping. All welding of connections shall be completed before the casing is hydrostatically tested (see 8.3.2).<br><br>6.5.2 • If specified, piping shall be gusseted in two orthogonal planes to increase the rigidity of the piped connection.<br><br>a) Gussets shall be of a material compatible with the pressure casing and the piping and shall be made of flat bar with a minimum cross section of 25 mm by 3 mm (1 in by 0.12 in).<br><br>b) Gussets shall be located at, or near the connection end of the piping, and fitted to the closest convenient location on the casing to provide maximum rigidity. The long width of the bar shall be perpendicular to the pipe and shall be located to avoid interference with the flange bolting or any maintenance areas on the pump.<br><br>c) Gusset welding must meet the fabrication requirements of 6.14.4, including PWHT when required, and the inspection requirements of this standard (see 8.2.2).<br><br>d) Gussets may also be bolted to the casing if drilling and tapping is done prior to the hydro test.<br><br>e) Piping may be clamped to gussets<br><br>6.5.3 If recommended by the supplier and approved by the purchaser, threaded |

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| | connections for pipe sizes exceeding DN 40 (1-1/2 NPS) may be used.<br><br>Note: For example:<br>a) on non-weldable materials, such as cast iron;<br>b) If essential for maintenance (disassembly and assembly).<br>c) where flanged or machined and studded openings are impractical<br><br>6.5.3.1 Unless otherwise recommended by the supplier and approved by the purchaser, pipe nipples screwed or welded to the casing should not be more than 150 mm (6 in.) long and shall be a minimum of Schedule 160 seamless for sizes DN 25 (NPS 1) and smaller and a minimum of Schedule 80 for DN 40 (NPS 1-1/2 ). |
| 2.3.3.1 A pipe nipple, preferably not more than 150 mm (6 inches) long, shall be screwed into the threaded opening.<br><br>2.3.3.2 Pipe nipples shall be a minimum of Schedule 160 seamless for NPS 1 and smaller and a minimum of Schedule 80 for NPS 11⁄2.<br><br>2.3.3.3 The pipe nipple shall be provided with a welding-neck or socketweld flange.<br><br>2.3.3.5 The threaded connection shall be seal welded; however, seal welding is not permitted on cast iron equipment, for instrument connections, or where disassembly is required for maintenance. Seal welded joints shall be in accordance with ASME B31.3.<br><br>2.3.3.4 The nipple and flange materials shall meet the requirements of 2.3.2. | 6.5.3.2 If recommended and approved, nipples longer than 150 mm (6 in.) shall be gusseted. |
| 2.3.3.6 Tapped openings and bosses for pipe threads shall conform to ASME B16.5.<br><br>2.3.3.7 Pipe threads shall be taper threads conforming to ASME B1.20.1. | 6.5.3.3 The pipe nipple shall be provided with a welding-neck or socket-weld flange |
| 2.3.4 Openings for NPS 1 1⁄4, 2 1⁄2, 3 1⁄2, 5, 7, and 9 shall not be used. | 6.5.3.4 All auxiliary connection to the pressure casing, except seal gland, shall terminate in a flange meeting the requirement of 6.6.1.1 or 9.6.1.2. These connections shall be integrally flanged, socket welded or butt welded as specified by the purchaser. Seal welding of threaded connection is not permitted. Purchase interface connection shall terminate in flange. |

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| | 6.5.3.5 • If specified, special threaded fittings for transitioning from the casing to tubing for seal flush piping may be used provided a secondary sealing feature such as O-rings are used and the joint does not depend on thread contact alone to seal fluid. The connection boss shall have a machined face suitable for sealing contact.<br><br>6.5.3.6 The nipple and flange shall meet the requirement of API 614/ISO 10438.<br><br>6.5.3.7 Unless otherwise specified, pipe thread shall be tapered threads conforming to ASME B1.20.1. Openings and bosses for pipe threads shall confirm to ASME B16.5<br><br>Note: For purposes of this provision, ASME B1.20.1 is equivalent to ISO 7-1. |
| 2.3.6.3 Flanges that are thicker or have a larger outside diameter than that required by ASME B16.5, API Standard 605 or MSS-SP-44 are acceptable.<br><br>2.3.6.4 Connections other than those covered by ASME B16.5, API Standard 605 or MSS-SP-44 require the purchaser's approval. When specified, the mating parts shall be furnished by the vendor. | 6.5.4 • If specified, cylindrical threads conforming to ISO 228-1shall be used. If cylindrical threads are used, they shall be sealed with a contained face gasket, and the connection boss shall have a machined face suitable for gasket containment. |
| 2.3.7 Machined and studded connections shall conform to the facing and drilling requirements of ASME B16.1, B16.5, or B16.42. Studs and nuts shall be furnished installed. The first 11⁄2 threads at both ends of each stud shall be removed. Connections larger than those covered by ASME shall meet the requirements of 2.3.6.4. | 6.5.4.1 • If specified, auxiliary connections to the pressure casing shall be machined and studded. These connections shall conform to the facing and drilling requirements of ASME B16.5 or ASME B16.1. Studs and nuts shall be furnished installed. The first 1½ threads at both ends of each stud shall be removed. Note: For the purpose of this provision, ASME B16.1 and ASME B16.5 are equivalent to ISO 7005-2 and ISO 7005-1, respectively.<br><br>6.5.4.2 All connections shall be suitable for the hydrostatic test pressure of the region of the casing to which they are attached. |
| 2.3.5 Tapped openings not connected to piping shall be plugged with solid round-head steel plugs furnished in accordance with ASME B16.11. As a minimum, these plugs shall meet the material requirements of the cylinder. Plugs that may later require removal shall be of corrosion-resistant material. Lubricant of the proper temperature specification shall be used on all threaded connections. Tape shall not be applied to threads of plugs inserted into oil passages. Plastic plugs are not permitted. | 6.5.4.3 Threaded openings not connected to piping shall be plugged. Taper-threaded plugs shall be long-shank solid round-head, or long-shank hexagon-head bar stock plugs in accordance with ASME B16.11. If cylindrical threads are specified, plugs shall be solid hexagon-head plugs in accordance with DIN 910. These plugs shall meet the material requirements of the casing. A lubricant that is suitable for the contained fluid and for the service temperature shall be used on all threaded connections. Thread tape shall not be used. Plastic plugs shall not be used. |

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| | **6.6 Flanges** |
| 2.3.6 Flanges shall conform to ASME B16.1, B16.5, or B16.42 as applicable, except as specified in 2.3.6.1 through 2.3.6.4. | 6.6.1 • Purchaser to specify whether ISO or ASME flanges are to be provided. |
| 2.3.6.1 Cast iron flanges shall be flat faced and shall have a minimum thickness of Class 250 per ANSI B16.1 for sizes 8 inches and smaller. | 6.6.1.1 Cast iron flanges shall be flat-faced and, except as noted in 6.4.2.4, conform to the dimensional requirements of ISO 7005-2 and the flange finish requirements of ASME B16.1 or ASME B16.42. PN 20 (Class 125) flanges shall have a minimum thickness equal to that of PN 40 (Class 250) flanges for sizes DN 200 (NPS 8) and smaller. Note 1: ISO 7005-2 (cast iron) flanges PN 20, 50 are designed to be interchangeable with ASME B16.1 (gray cast iron) and B 16.42 (ductile cast iron) but they are not identical. They are deemed to comply with dimensions specified in ASME B16.1 (gray cast iron) and B 16.42 (ductile cast iron).<br><br>6.6.1.2 Flanges other than cast iron shall, as a minimum requirement, conform to the dimensional requirements of ISO 7005-1 PN 50 except as noted in 6.4.2.4 and the flange finish requirements of ASME B16.5 or ASME B16.47.<br><br>Note 1: For the purpose of this provision, ASME B16.5 Class 300, ASME B16.47 Class 300 and EN 1759-1 class 300 are equivalent to ISO 7005-1 PN 50.<br><br>Note 2: ISO 7005-1 (steel flanges) PN 20, 50, 110, 150, 260, 420 are designed to be interchangeable with ASME B16.5 and MSS SP-44 flanges - ISO 7005-1 flanges are not identical to ASME B 16.5 and MSS SP 44 flanges but are deemed to comply with the dimensions specified in the ASME B 16.5 and MSS SP 44. |
| 2.3.6.2 Flat-faced flanges with full raised-face thickness are acceptable on cases other than cast iron, with purchaser's approval. | 6.6.1.3 Flanges in all materials that are thicker or have a larger outside diameter than required by the relevant ISO (ASME) standards in this standard are acceptable. Non-standard (oversized) flanges shall be identified as such and completely dimensioned on the arrangement drawing. If oversized flanges require studs or bolts of non-standard length, this requirement shall be identified as such on the arrangement drawing.<br><br>6.6.1.4 Flanges shall be full-faced or spot-faced on the back and shall be designed for through-bolting, except for jacketed casings.<br><br>6.6.1.5 Unless otherwise specified, the supplier shall provide mating flanges, studs and |

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| | nuts for non-standard connections.<br><br>6.6.1.6 Studs or bolt holes shall straddle centerlines parallel to the main axes of the equipment. |
| 2.3.8 All of the purchaser's connections shall be accessible for disassembly without the machine being moved. | 6.6.1.7 All of the purchaser's connections shall be accessible for disassembly without requiring the machine, or any major part of the machine, to be moved. |
| **2.4 External Forces and Moments** | **6.7 External Forces and Moments** |
| For pumps with steel or alloy-steel casings, inlet and outlet nozzles shall be capable of withstanding forces and moments from external piping determined by the following formulas (see Figure 1):<br>Fx = 13D Mx = 7D<br>Fy = 13D My = 7D<br>Fz = 13D Mz = 7D<br>or in conventional units<br>Fx = 75D Mx = 125D<br>Fy = 75D My = 125D<br>Fz = 75D Mz = 125D | 6.7.1 The vendor shall specify, in the quotation, the magnitude of forces and moments which may be applied, simultaneously, to the inlet and outlet connections at the rated operating conditions. As a minimum, the pump inlet and outlet connection must be capable of withstanding the limits indicated in Table 2<br><br>For pumps with connection sizes not listed in Table 2, the inlet and outlet nozzles shall be capable of withstanding forces and moments from external piping determined by the following formulas<br>Fx = 13D Mx = 7D<br>Fy = 13D My = 7D<br>Fz = 13D Mz = 7D |

| SI Units Nominal Size of Flange (DN) | Forces each nozzle Fx, Fy and Fz (N) | Moments each nozzle Mx, My and Mz (N-m) |
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| <=50 | 650 | 350 |
| 80 | 1040 | 560 |
| 100 | 1300 | 700 |
| 150 | 1950 | 1050 |
| 200 | 2600 | 1400 |
| 250 | 3250 | 1750 |
| 300 | 3900 | 2100 |
| 350 | 4550 | 2450 |
| 400 | 5200 | 2800 |
| 500 | 6500 | 3500 |
| 600 | 7800 | 4200 |

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| | or in conventional units<br>Fx = 75D Mx = 125D<br>Fy = 75D My = 125D<br>Fz = 75D Mz = 125D |
| Where:<br>D = nominal pipe size of the pump nozzle connection in millimeters (inches).<br>Fx = force in Newtons (pounds) on the x-axis, which is parallel to the shaft axis.<br>Fy = horizontal force in Newtons (pounds) on the yaxis, which is mutually perpendicular to the x- and z-axis.<br>Fz = vertical force in Newtons (pounds) on the z-axis, which is mutually perpendicular to the y- and x-axes.<br>Mx = moments around the x-axis, in Newton-meters (pound-feet).<br>My = moments around the y-axis, in Newton-meters (pound-feet).<br>Mz = moments around the z-axis, in Newton-meters (pound-feet).<br>The vendor shall submit comparable criteria for pump casings constructed of other materials. | Where:<br>D = nominal pipe size of the pump nozzle connection in millimeters (inches).<br>Fx = force in Newtons (pounds) on the x-axis, which is parallel to the shaft axis.<br>Fy = horizontal force in Newtons (pounds) on the y-axis, which is mutually perpendicular to the x- and z-axis.<br>Fz = vertical force in Newtons (pounds) on the z-axis, which is mutually perpendicular to the y- and x-axes.<br>Mx = moments around the x-axis, in Newton-meters (pound-feet).<br>My = moments around the y-axis, in Newton-meters (pound-feet).<br>Mz = moments around the z-axis, in Newton-meters (pound-feet).<br>The vendor shall submit comparable criteria for pump casings constructed of other materials. |
| | 6.7.2 Casing Liners |
| | 6.7.2 1 •if specified, replaceable liners shall be provided for Screw pumps. |
| | 6.7.2.2 Unless otherwise specified for multiphase twin screw pumps, hard coatings and/or surface hardening shall be applied to the liner bores |
| | Note Hard-coated and/or surface hardened liners are used to reduce the rate of degradation due to abrasion. See Annex B for additional information. |
| 2.5 Rotating Elements | **6.8 Rotating Elements** |
| | 6.8.1 Rotors |
| 2.5.1 Stationary and moving pumping elements shall be designed and fabricated of material to prevent galling. Rotating parts shall be properly aligned. Internal loads shall be fully supported by the use of such means as hydraulic balance, bearings, or bushings. | 6.8.1.1 Stationary and moving pumping elements shall be designed and fabricated of material to prevent galling. Rotating parts shall be properly aligned. Internal loads shall be fully supported by the use of such means as hydraulic balance, bearings, or bushings. |

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| 2.5.2 Rotors and shafts shall be of sufficient stiffness and material compatibility to prevent wear between the rotor bodies and the casing, and between gear-timed rotor bodies at the most unfavorable specified conditions, including 110 percent of the relief-valve set pressure. Rotor bodies not integral with the shaft shall be permanently fixed to the shaft to prevent unintended relative motion under any operating condition. Rotors and shafts shall be of materials that have corrosion and erosion resistance compatible with the application. | • 6.8.1.2 If specified, for twin screw pumps, rotor stiffness shall be adequate to prevent contact between the rotor bodies and the casing and between gear-timed rotor bodies at the most unfavorable specified conditions.<br><br>6.8.1.3 For twin screw multiphase pumps (MPPs), the maximum allowable rotor deflection under the worst operating condition (consider temperature, MAWP, nozzle loads, particulate etc as specified on the datasheet) shall be calculated and able to be demonstrated through computer modeling to show non-contact between rotors and the surrounding pump casing.<br><br>Note 1: If requested, the purchased shall be able to review these calculations<br><br>Note 2: This requirement does not apply to PCPs, since continuous contact/interference is required by design between<br><br>6.8.1.4 Rotor bodies not integral with the shaft shall be positively attached to the shaft to prevent relative motion under any condition. Structural welds on rotors shall be full-penetration continuous welds and shall be stress relieved with appropriate ASTM heat treatment procedure.<br><br>6.8.1.5 For multiple screw pumps, each rotor set shall be clearly marked with a unique identification number on each male and female rotor. This number shall be on the end of the shaft opposite the coupling or in an accessible area that is not prone to maintenance damage.<br><br>6.8.1.6 All shaft keyways shall have fillet radii conforming to ANSI/ASME B17.1.<br><br>• 6.8.1.7 If specified, hardened rotor shall be provided for screw pumps.<br><br>6.8.1.8 Unless otherwise specified for MPPs, hard coatings and/or shall be applied to the rotors.<br><br>Note Hard-coated and/or surface hardened liners are used to reduce the rate of degradation due to abrasion. See Annex B for additional information.<br><br>6.8.1.9 Rotors for twin screw pumps shall be dynamically balanced to ISO 1940-1 grade |

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| | 6.8.2 Timing gears |
| 2.5.3 Timing gears, when furnished, may be spur, helical, or herringbone type. All gears shall be the coarse-pitch type, and the gear quality shall be at least 9, as defined by AGMA. The gears shall be designed in accordance with AGMA Standard 6010 with a minimum service factor of 1.5. | 6.8.2.1 Timing gears, when furnished may be spur, helical, or herringbone type. All gears shall be the coarse-pitch type, and the gear quality shall be at least 9 as defined by AGMA. The gears shall be designed in accordance with AGMA Standard 6010 with a minimum service factor of 1.5. |
| | 6.8.2.2 For twin screw MPPs, the timing gears shall be designed in accordance with AGMA 6010 with a minimum service factor of 2.0. The timing gears shall be manufactured to minimum quality level of 11 as defined by AGMA |
| **2.6 Mechanical Seals And Conventional Packing** | **6.9 Mechanical Shaft Seals** |
| 2.6.1 Mechanical Seals | |
| 2.6.1.1 Unless otherwise specified, mechanical seals shall be furnished. | 6.9.1 Unless otherwise specified, cartridge mechanical shaft seals shall be furnished. |
| 2.6.1.2 Mechanical seals shall be of the single-balanced type (one rotating face per seal chamber) with either a sliding gasket or a bellows between the axially movable face and the shaft sleeve or housing. Unbalanced seals shall be furnished when specified or approved by the purchaser. Other special configurations may be specified, or they may be recommended by the vendor if required for the service. Double seals have two rotating faces per chamber, sealing in opposite directions.<br><br>Tandem seals have two rotating faces per chamber, sealing in the same direction. | 6.9.2 Seal selection shall be suitable for specified variations in suction and/or discharge conditions during start-up, operation, or shutdown, including possible upset conditions.<br><br>Note: If the seal is exposed to suction pressure, special consideration may also be necessary for low suction pressure conditions or when a pump is subjected to a NPSH test requirement.<br><br>6.9.3 Unless otherwise specified, shaft seals shall be provided in accordance with API 682/ISO 21049.<br><br>6.9.4 ● If specified, manufacturer's standard seal meeting the intent of API 682/ISO 21049 shall be supplied. The following items shall be provided in the proposal:<br>a) category<br>b) type<br>c) arrangement and geometry<br>d) materials of construction<br>e) reference list |

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| | Note: Space or design parameters for some pump types, sizes, or applications make the use of API 682/ISO 21049 seals impractical. |
| 2.6.1.3 This standard does not cover the design of the component parts of mechanical seals; however, the design and materials of the component parts shall be suitable for the specified service conditions. The components shall also withstand the maximum discharge pressure except in high-discharge-pressure service where this requirement is impractical. For such applications, the vendor shall advise the purchaser of the maximum sealing pressure and the maximum dynamic and static pressure ratings of the seal. | 6.9.5 This standard does not cover the design of the component parts of mechanical seals. However, the design of the component parts shall be suitable for the specified service conditions and consistent with API 682/ISO 21049. The purchaser shall specify the seal requirements using the selection process and the data sheets in API 682/ISO 21049. |
| | 6.9.6 Single seals must be equipped with a close fitting throttle bushing on the atmospheric side of the seal to restrict the rate of leakage. If this is not possible due to space limitations, a suitable means of detecting and controlling the leakage shall be provided. |
| 2.6.1.4 Mechanical-seal materials shall be furnished in accordance with Appendix F. Seal faces and gaskets shall be coded in accordance with Tables F-1 and F-2 | 6.9.7 The seal shall be accessible for inspection and removal without disturbing the driver. Unless otherwise specified, a spacer coupling, with a spacer of the next standard length longer than the seal shall be provided by the supplier. |
| 2.6.1.5 When a seal gland is used, its component parts shall be satisfactory for the maximum seal-chamber design pressure and pumping temperature. It shall have sufficient rigidity to avoid any distortion that would impair seal operation, including distortion that may occur during tightening of the bolts to set gasketing. | 6.9.8 Seal chamber face run-out (TIR) shall not exceed 0.5 microns/mm (0.0005”/in) of seal chamber bore diameter.<br><br>6.9.9 If a seal gland is used, its component parts shall be satisfactory for the maximum seal-chamber design pressure and pumping temperature. It shall have sufficient rigidity to avoid any distortion that would impair seal operation, including distortion that may occur during tightening of the bolts.<br><br>6.9.10 The mating joint between the seal gland and the seal chamber face shall incorporate a confined gasket to prevent blowout. The gasket shall be of the controlled compression type (o-ring or spiral-wound gasket) with metal to metal gland to seal-chamber contact. If space or design parameters make this requirement impractical, an alternative seal gland design shall be submitted to the purchaser for approval.<br><br>6.9.11 Specified seal and pump connections shall be identified by symbols permanently marked on the component. Symbols shall be in accordance with those specified in API |

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| | 682/ISO 21049. If a seal gland utilizes a common design with multiple ports for symmetrical installation on multiple shaft pumps (right and left), the ports shall be appropriately identified. |
| 2.6.1.6 The seal chamber shall be provided with an internal or external vent to permit complete venting of the chamber before startup. | 6.9.12 Seal glands and seal chambers shall have provision for only those connections required by the seal flush plan. If other tapped connections are present but not used, they shall be plugged and labeled in accordance with API 682/ISO 21049. |
| 2.6.1.7 If seal flushing and cooling is provided by the pumped fluid, the pump vendor shall ensure that sufficient flow reaches the primary seal faces to provide for cooling and maintenance of a stable film at the seal faces. | 6.9.13 The seal chamber shall be provided with an internal or external vent to permit complete venting of the chamber before startup. |
| If the purchaser provides an external source of seal flushing, the pump vendor shall specify the flow, pressure, temperature, and required lubricating properties of the flushing medium. If a restriction orifice is used, it shall not be less than 3 mm (1⁄8 inch) in diameter. | 6.9.14 If seal flushing and cooling is provided by the pumped fluid, the pump supplier shall ensure that sufficient flow reaches the primary seal faces to provide for cooling and maintenance of a stable film at the seal faces. Allowance for cooling flow must be made when determining the pump capacity to meet delivered flow requirements.<br><br>6.9.15 If an external source of seal flushing is provided by the purchaser, the pump supplier shall specify the flow, pressure, temperature, and required lubricating properties of the flushing medium. If a restriction orifice is used, it shall not be less than 3 mm (1⁄8 inch) in diameter. |
| | 6.9.16 ● If specified, jackets shall be provided on seal chambers for heating. Heating requirements shall be agreed upon by the purchaser, pump manufacturer, and seal manufacturer. |
| 2.6.1.8 Where leakage past the primary mechanical seal must be contained, additional shaft sealing devices may be specified. These devices typically include double or tandem mechanical seals, auxiliary stuffing boxes, or secondary seals. The purchaser and the vendor shall mutually agree on the manner of lubricating these devices. | 6.9.17 If cooling is required, unless otherwise specified air cooling shall be provided. |
| 2.6.1.9 Unless otherwise specified, mechanical seals shall be installed in the pump prior to shipment and shall be clean, lubricated, and ready for service. | 6.9.18 Unless otherwise specified, mechanical seals shall be installed in the pump prior to shipment and shall be clean, lubricated, and ready for service. If seals require final adjustment or installation in the field, a metal tag shall be attached warning of this requirement. |

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| | ● 6.9.19 If specified, in dual seal applications, the maximum seal leakage to atmosphere at the specified operating conditions, and the expected inner seal-oil leakage rates, if applicable, shall be provided.<br><br>Note: This information is required to determine the rate of barrier or buffer seal oil usage and thus the sizing of the seal oil reservoir. |
| 2.6.2 Stuffing Boxes for Conventional Packing<br><br>2.6.2.1 Unless otherwise specified, the packing type and material shall be selected by the vendor for the specified service.<br><br>2.6.2.2 Stuffing boxes shall have not less than four rings of packing plus a lantern ring. The stuffing boxes shall preferably accommodate five rings of packing plus a lantern ring. The minimum packing size permitted shall be 6mm (1⁄4 inch) square. The packing size shall preferably be a minimum of 9mm (3⁄8 inch) square.<br><br>2.6.2.3 When specified, or when recommended by the vendor, a lantern ring with a width at least 11⁄2 times the packing size shall be provided for the introduction of cooling and/or lubricating media directly into the packing. Connections to the lantern ring shall be a minimum of NPS 1⁄4.<br><br>2.6.2.4 Ample space shall be provided for packing replacement without removing or dismantling any part other than the gland.<br><br>2.6.2.5 Unless otherwise specified, packing shall be installed by the pump vendor before shipment.<br><br>2.6.2.6 Glands on all oil pumps shall be designed so that the bolts cannot slip if the packing becomes loose. | |
| **2.7 Bearings**<br><br>2.7.1 Antifriction bearings shall have a minimum L-10 rated life (see AFBMA Standard 9) of either 25,000 hours with continuous operation at rated conditions, or 16,000 hours at maximum axial and radial loads and rated speed. | **6.10 Bearings and Bearing Housings**<br><br>6.10.1 Antifriction bearings shall have a minimum L-10 rated life (see AFBMA Standard 9) of either 25,000 hours with continuous operation at rated conditions, or 16,000 hours at maximum axial and radial loads and rated speed. These L-10 lives shall also apply for belt |

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| | driven pumps. |
| Note: The rated life is the number of hours at rated bearing load and speed that 90 percent of the group of identical bearings will complete or exceed before the first evidence of failure. It is recognized that this life may not be achieved where bearings are operated in fluids other than clean lubricating oil. | Note 1: The rated life is the number of hours at rated bearing load and speed that 90 % of the group of identical bearings will complete or exceed before the first evidence of failure. It is recognized that this life may not be achieved where bearings are operated in fluids other than clean lubricating oil.<br><br>Note 2: In twin screw MPPs, the bearings must be sized and selected to manage the full radial load applied to the shaft, since hydrodynamic fluid film support between the rotor and liner may not be available. Special attention should also be given to minimum loading conditions to ensure that bearing skidding does not occur. |
| 2.7.2 Antifriction bearings shall be retained on the shaft and fitted into housings in accordance with the requirements of AFBMA Standard 7; however, the device used to lock ball thrust bearings to the shaft shall be restricted by a nut with a tongue-type lock washer, for example, Series W. | |
| 2.7.3 Except for the angular contact type, antifriction bearings shall have a loose internal clearance fit equivalent to AFBMA Symbol 3, as defined in AFBMA Standard 20. Tapered roller bearings shall have a clearance fit as described in AFBMA 11. Single- or double-row bearings shall be of the Conrad type (no filling slots). | 6.10.2 Except for the angular contact type, antifriction bearings shall have a loose internal clearance fit equivalent to AFBMA Symbol 3, as defined in AFBMA Standard 20. Tapered roller bearings shall have a clearance fit as described in AFBMA 11. Single or double-row bearings shall not be supplied with filling slots.<br><br>Note 1: Greater internal clearance may reduce the temperature rise of the lubricant. However, vibration velocities may be increased with greater clearances.<br><br>Note 2: For the purpose of this provision, AFBMA 20 Group 3 is equivalent to ISO 5753 Group 3 |
| 2.7.4 Housings for separately lubricated bearings shall be sealed against external contaminants. Such housings for oil-lubricated bearings shall contain a drain at the low point and shall be equipped with an oil-level gauge. | 6.10.3 Housings for separately lubricated bearings shall be sealed against external contaminants. Such housings for oil-lubricated bearings shall contain a drain at the low point and shall be equipped with an oil-level gauge. |
| 2.7.5 When regreaseable-type lubricated bearings are supplied, the manufacturer's design shall include a provision to protect against over greasing. | 6.10.4 If re-greaseable type lubricated bearings are supplied, the manufacturer's design shall include a provision to protect against over greasing.<br><br>6.10.5 Rolling element bearings shall be located, retained and mounted in accordance |

with the following.

6.10.5.1 Bearings shall be retained on the shaft with an interference fit and fitted into the housing with a diametral clearance, both in accordance with the recommendations of AFBMA Standard 7, or as recommended by the bearing manufacturer.

6.10.5.2 Bearings shall be mounted directly on the shaft. Bearing carriers are acceptable only with customer approval.

6.10.5.3 Bearings shall be located on the shaft using shoulders, collars or other positive locating devices; snap rings and spring-type washers are not acceptable.

Note: This paragraph applies to all rolling element bearings, including both ball and roller types. For certain roller bearings, such as cylindrical roller types with separable races, bearing housing diametral clearance may not be appropriate.

**6.11 Vibration Limits**

6.11.1 Vibration Limits for Liquid Pumps

| Measurement on Bearing Housing | Rolling Element Bearings |
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| Steady State Vibration at any speed within operating range on test in the field | Vu<3.8 mm/s RMS (0.15 in/s RMS) |

6.11.2 Vibration Limits for Multiphase Pumps (MPPs)

| Measurement on Bearing Housing | Rolling Element Bearings |
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| For full liquid or any GVF on test | Vu<7.1 mm/s RMS (0.28 in/s RMS) |
| Steady State Vibration at any speed within operating range in the field | Vu<5.5 mm/s RMS (0.22 in/s RMS) |

Where: Vu – unfiltered velocity
RMS = Root Mean Squared

**2.8 Lubrication**

**6.12Lubrication**

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| | 6.12.1 Lubrication for rotary pumps<br><br>6.12.2 If internal bearings and/or internal timing gears are used the pump supplier shall verify that the pumped fluid will provide suitable lubrication. |
| 2.8.1 Unless otherwise specified, bearings and bearing housings shall be arranged for hydrocarbon oil lubrication. | 6.12.3 Unless otherwise specified, bearings and bearing housings shall be splash lubricated and designed for mineral (hydrocarbon) oil. |
| 2.8.2 The vendor shall specify the type, amount, and frequency of lubrication for separately lubricated bearings and timing gears. | 6.12.4 Sufficient cooling, including an allowance for fouling shall be provided to maintain bearing and oil life. |
| 2.8.3 When specified, oil-lubricated bearings in separate housings shall be furnished with constant level oilers. | 6.12.5 Based on the specified operating conditions and an ambient temperature of 43$^{0}$C (110 $^{0}$F), Oil lubrication temperature shall be in accordance with 6.12.11.4. |
| 2.8.4 Any points that require grease lubrication shall have suitable extension lines to permit access during operation. | 6.12.5.1 For Pressurized systems oil outlet temperature below 70 $^{0}$ C (160 $^{0}$F) and bearing metal temperature (if bearing-temperature sensors are supplied) less than 93 $^{0}$C (200 $^{0}$ F).<br><br>6.12.5.2 During shop testing, and under the most adverse specified operating conditions, the bearing-oil temperature rise shall not exceed 25K (50 $^{0}$ R).<br><br>6.12.5.3 For ring-oiled or splash systems, oil sump temperature must be maintained below 82 $^{0}$C (180 $^{0}$F). During shop testing, the sump oil temperature rise shall not exceed 40k (70 $^{0}$R) and (if bearing-temperature sensors are supplied) outer ring temperature shall not exceed 93 $^{0}$C (200 $^{0}$F).<br><br>NOTE Pump equipped with ring-oiled or splash lubrication system might reach temperature stabilization during performance test of short duration.<br><br>6.12.6 If specified, or as recommended by the pump vendor, the bearing lubrication may be splash, positive pressure, or gravity lubricated. A sight glass gauge or oil-level dipstick shall be provided.<br><br>6.12.7 Unless otherwise specified, pressurized oil system shall confirm to the requirements of the “General Purpose” section ISO 10438-3.<br><br>NOTE For the purpose of this provision, Chapter 3 of API 614 is equivalent ISO 10438-3. |

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| | 6.12.8 IF specified or if recommended by the vendor and approved by the purchaser, a pressure lubrication system shall be provided to supply oil at a suitable pressure and temperature to the pump, the driver, and any other driven equipment, including gears.<br><br>6.12.9 External pressure lubrication system shall comply with the requirements of ISO10438-3.<br><br>NOTE For the purpose of this provision, Chapter 3 of API 614 is equivalent to ISO 10438-3.<br><br>6.12.10 If specified, the purpose lubrication system shall confirm to the requirements of ISO 10438-2 (Special-purpose oil system). For such a lubrication system, datasheets should be supplied.<br><br>NOTE For the purpose of this provision, Chapter 2 of API 614 is equivalent to ISO 10438-2.<br><br>6.12.11 If grease lubricated rolling-element bearing are specified, lubrication shall be accordance with 6.12.11.1 through 6.12.11.4.<br><br>6.12.11.1 Grease life (re lubrication interval) shall be using the method recommended by the bearing manufacturer or an alternative method approved by the purchaser.<br><br>6.12.11.2 Grease lubrication shall not be estimated grease life is less then 2000 hours.<br><br>6.12.11.3 If the estimated grease life is 2000 hours or greater, but less than 25,000 hours, provision shall, be made for re-greasing the bearing in service and for the effective discharge of old excess grease, and the vendor shall advise the purchaser of the required re-greasing interval.<br><br>6.12.11.4 If the grease life is 25,000 hours or more, grease nipples or any other system for the addition of grease in-service shall not be fitted. |
| **2.9 Materials** | **6.13 Materials** |

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| 2.9.1 General<br><br>2.9.1.1 Materials of construction shall be manufacturer's standard for the specified operating conditions, except as required or prohibited by the data sheets or this standard. (See 3.5 for auxiliary piping material requirements.) The metallurgy of all major components shall be clearly stated in the vendor's proposal. | **6.13.1 Material Inspection of Pressure-Containing Parts**<br><br>6.13.1.1 Regardless of the generalized limits presented in this section, it shall be the vendor's responsibility to review the design limits of all materials and welds in the event that more stringent requirements are required.<br><br>6.13.1.2 Defects that exceed the limits imposed in 6.13.3 and 16.3.4 shall be removed to meet the quality standards cited, as determined by additional magnetic particle or liquid penetrant inspection as applicable before repair welding.<br><br>6.13.1.3 The purchaser shall be notified before making a major repair to a pressure containing part. Major repairs, for the purpose of purchaser notification only, is any defect that equals or exceeds any of the three criteria defined below<br><br>a) The depth of the cavity prepared for repair welding exceeds 50% of the component wall thickness.<br><br>b) The length of the cavity prepared for repair welding is longer than 150 mm (6 in.) in any direction.<br><br>c) The total area of all repairs to the part under repair exceeds 10% of the surface area of the part.<br><br>6.13.1.4 All repairs to pressure containing parts shall be made as required by the following documents:<br><br>a) The repair of plates, prior to fabrication, shall be performed in accordance with the ASTM standard to which the plate was purchased.<br><br>b) The repair of castings or forgings shall be performed prior to final machining in accordance with the ASTM standard to which the casting or forging was purchased.<br><br>c) The inspection of a repair of a fabricated casing or the defect in either a weld or the base metal of a cast or fabricated casing, uncovered during preliminary or final machining, shall be performed in accordance with 8.2.2.1.1. |

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| | **6.13.2 Materials of Construction**<br><br>6.13.2.1 Unless otherwise specified the materials of construction of the pressure-containing casing shall be carbon steel as a minimum.<br><br>6.13.2.2 The materials shall be the vendor's standard for the operating conditions specified, except as required by the data sheet or this standard. |
| 2.9.1.2 Materials shall be identified in the proposal with the applicable ASTM, AISI13, ASME, or SAE14 numbers, including material grade (See Appendix B). When no such designation is available, the vendor's material specification, giving physical properties, chemical composition, and test requirements, shall be included in the proposal. | 6.14.2.3 The materials of construction of all major components shall be clearly stated in the vendor's proposal. Materials shall be identified by reference to applicable international standards, including the material grade. If no such designation is available, the vendor's material specification, giving physical properties, chemical composition, and test requirements shall be included in the proposal.<br><br>• 6.13.2.4 If specified, copper or copper alloys shall not be used for parts which are in contact with process fluids. Nickel-copper alloy (UNS NW 4400 or UNS N04400), bearing babbitt, and copper-containing precipitation-hardened stainless steels are excluded from this requirement.<br><br>*Warning: Certain corrosive fluids in contact with copper alloys have been known to form explosive compounds.* |
| 2.9.1.3 The vendor shall specify the ASTM optional tests and inspection procedures that may be necessary to ensure that materials are satisfactory for the service. Such tests and inspections shall be listed in the proposal. The purchaser may consider specifying additional tests and inspections, especially for materials used in critical components. | 6.13.2.5 The vendor's response to the inquiry shall specify the optional tests and inspection procedures that are necessary to ensure that materials are satisfactory for the service. Such tests and inspections shall be listed in the proposal.<br><br>Note: The purchaser may specify additional optional tests and inspections, especially for materials used for critical components or in critical services. |
| 2.9.1.4 External parts subject to rotary or sliding motions (such as control linkage joints and adjusting mechanisms) shall be of corrosion resistant materials suitable for the site environment. | 6.13.2.6 External parts that are subject to rotary or sliding motions (such as control linkage joints and adjustment mechanisms) shall be of corrosion resistant materials suitable for the site environment. |
| 2.9.1.5 Minor parts that are not identified (such as nuts, springs, washers, gaskets, and keys) shall have corrosion resistance at least equal to that of specified parts in the same | 6.13.2.7 Minor parts such as nuts, springs, washers, gaskets, and keys shall have corrosion resistance at least equal to that of specified parts in the same environment. |

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| environment. | |
| 2.9.1.6 The purchaser will specify any corrosive agents in the motive and process fluid and in the environment, including constituents that may cause stress corrosion cracking. | • 6.13.2.8 The purchaser shall specify any erosive or corrosive agents (including trace quantities) present in the process fluids and in the site environment, including constituents that may cause stress-corrosion cracking or attack elastomers.<br><br>Note: Typical agents of concern are hydrogen sulfide, amines, chlorides, bromides, iodides, cyanides, fluorides, naphthenic acid and polythionic acid. Other agents affecting elastomer selection include ketones, ethylene oxide, sodium hydroxide, methanol, benzene and solvents. |
| 2.9.1.7 If parts exposed to conditions that promote inter-granular corrosion are to be fabricated, hard faced, overlaid, or repaired by welding, they shall be made of low-carbon or stabilized grades of austenitic stainless steel.<br><br>Note: Overlays or hard surfaces that contain more than 0.10 percent carbon can sensitize both low-carbon and stabilized grades of austenitic stainless steel unless a buffer layer that is not sensitive to inter granular corrosion is applied. | 6.13.2.9 If austenitic stainless steel parts exposed to conditions that may promote inter-granular corrosion are to be fabricated, hard-faced, overlaid or repaired by welding, they shall be made of low-carbon or stabilized grades.<br><br>Note: Overlays or hard surfaces that contain more than 0.10% carbon can sensitize both low-carbon and stabilized grades of austenitic stainless steel unless a buffer layer that is not sensitive to inter-granular corrosion is applied.<br><br>• 6.13.2.10 If specified, the vendor shall furnish material certificates that include chemical analysis and mechanical properties for the heats from which the material is supplied for pressure-containing castings and forgings and shafts. Unless otherwise specified, piping nipples, auxiliary piping components, and bolting are excluded from this requirement. |
| 2.9.1.8 Where mating parts such as studs and nuts of AISI Standard Type 300 stainless steel or materials with similar galling tendencies are used, they shall be lubricated with an anti seizure compound of the proper temperature specification and compatible with the specified fluids.<br><br>Note: It is preferable to use materials which do not have galling tendencies. Also, torque loading values will be considerably different with and without anti seizure compound. | 6.13.2.11 If mating parts such as studs and nuts of austenitic stainless steel or materials with similar galling tendencies are used, they shall be lubricated with an anti-seizure compound of the proper temperature specification and compatible with the specified process liquid.<br><br>Note: The torque loading values to achieve the necessary preload are likely to vary considerably depending upon whether or not an anti-seizure compound is used. |
| 2.9.1.9 Materials exposed to a sour environment (wet H2S) as defined by NACE MR-01-90 shall be in accordance with the requirements of that standard. Ferrous materials not covered by NACE MR-01-90 shall be limited to a yield strength not exceeding 6200 bar (90,000 psi) and a hardness not exceeding Rockwell C22. | 6.13.2.12 The purchaser shall specify the amount of wet H2S that may be present, considering normal operation, start-up, shutdown, standby, upsets or unusual operating conditions such as catalyst regeneration. |

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| Note: It is the responsibility of the purchaser to determine the amount of H2S that may be present, considering normal operation, startup, shutdown, idle standby, upsets, or unusual operating conditions such as catalyst regeneration. In many applications, small amounts of H2S are sufficient to require NACE materials. When there are trace quantities of H2S known to be present or if there is any uncertainty about the amount of H2S that may be present, the purchaser should automatically note on the data sheets that NACE materials are required.<br>Components that are fabricated by welding shall be stress relieved, if required, so that both the welds and the heat-affected zones meet the yield strength and hardness requirements. The purchaser will specify the presence of such agents in the media.<br><br>2.9.1.10 When dissimilar materials with significantly different electrical potentials are placed in contact with the presence of an electrolytic solution, galvanic couples that can result in serious corrosion of the less noble material may be created. If such conditions exist, the purchaser and the vendor should select materials in accordance with the NACE Corrosion Engineer's Reference Book.<br><br>2.9.1.12 The use of ASTM A 515 steel is prohibited. Low-carbon steels can be notch-sensitive and susceptible to brittle fracture at ambient or low temperatures. Therefore, only fully killed, normalized steels made to fine-grain practice are acceptable.<br><br>2.9.1.11 Materials, casting factors, and the quality of any welding shall be equal to those required by Section VIII, Division 1, of the ASME Code. The manufacturer's data report forms, as specified in the code, are not required. | NOTE: In many applications, small amounts of wet H2S are sufficient to require materials resistant to sulfide stress-corrosion cracking. If there are trace quantities of wet H2S known to be present or if there is any uncertainty about the amount of wet H2S that may be present, the purchaser should consider specifying that reduced hardness materials are required.<br><br>6.13.2.13 The purchaser shall specify if reduced hardness materials are required.<br><br>6.13.2.13.1 If reduced harness materials are specified in 6.13.2.13, they shall be supplied in accordance with NACE MR 0103 or MR 0175 (ISO 15156)<br><br>Note 1: NACE MR 0103 applies to oil refineries, LNG plants and chemical plants. NACE MR 0103 applies to materials potentially subject to corrosion cracking.<br><br>Note 2: NACE MR 0175 applies to oil and gas production facilities and natural gas sweetening plants. NACE MR 0175 applies to materials potentially subject to corrosion cracking. NACE MR 0175 is equivalent to ISO 15156<br><br>6.13.2.13.2 If reduced hardness materials are specified, ferrous materials not covered by NACE MR 0103 or MR 0175 (ISO 15156) shall have a yield strength not exceeding 620 N/mm2 (90000 psi) and a hardness not exceeding HRC 22. Components that are fabricated by welding shall be post-weld heat treated, if required, so that both the welds and the heat-affected zones meet the yield strength and hardness requirements.<br><br>6.13.2.13.3 If reduced hardness materials are specified, the following components shall have reduced hardness<br>a) the pressure casings<br>b) Shafting (including wetted shaft nuts)<br>c) pressure retaining mechanical seal components (excluding seal face)<br>d) wetted bolting<br><br>6.13.2.14 The vendor shall select materials to avoid conditions that may result in electrolytic corrosion. If such conditions cannot be avoided, the purchaser and the vendor shall agree on the material selection and any other precautions necessary.<br><br>Note: If dissimilar materials with significantly different electrical potentials are placed in |

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| | contact in the presence of an electrolytic solution, galvanic couples that can result in serious corrosion of the less noble material may be created. That can result in serious corrosion of the less noble material. The NACE Corrosion Engineer's Reference Book is one resource for selection of suitable materials in these situations.<br><br>6.13.2.15 Steel made to a coarse austenitic grain size particle such as (ASTM A515) shall not be used. Only fully killed or normalized steels made to fine grain practice shall be used.<br><br>6.13.2.16 The manufacturer's data report forms, as specified in codes such as ASME BPVC Section VIII, are not required.<br><br>Note: For impact requirements refer to 6.13.6.4.<br><br>• 6.13.2.17 The material specification of all gaskets and O-rings exposed to the pumped fluid shall be identified in the proposal. O-rings shall be selected and their application limited in accordance with ISO 21049. O-ring materials shall be compatible with all specified services. Special consideration shall be given to the selection of O-rings for high-pressure services to ensure that they will not be damaged upon rapid Depressurization (explosive decompression). It shall be specified on the datasheet if the service is such that there is a risk of rapid depressurization.<br><br>Note 1: For the purpose of this provision, API Standard 682-2002 is equivalent to ISO 21049.<br><br>Note 2: Susceptibility to explosive decompression depends on the gas to which the O-ring is exposed, the compounding of the elastomer, temperature of exposure, the rate of decompression, and the number of cycles. |
| 2.9.1.13 The minimum quality bolting material for pressure joints for cast iron casings shall be carbon steel (ASTM A 307, Grade B); and for steel casings shall be high-temperature alloy steel (ASTM A 193, Grade B7). Nuts shall conform to ASTM A 194, Grade 2H (or ASTM A 307, Grade B, case hardened, where space is limited). For temperatures below -30°C (-20°F), low-temperature bolting material in accordance with ASTM A 320 shall be used. | 6.13.2.18 The minimum quality bolting material for pressure-retaining parts shall be carbon steel (such as ASTM A 307, Grade B) for cast iron casings; and high temperature alloy steel (such as ASTM A 193, Grade B7) for steel casings. Carbon steel nuts (such as ASTM A 194, Grade 2H) shall be used, except that case hardened carbon steel nuts (such as ASTM A 563, Grade A) shall be used where space is limited. For temperatures below –29°C (–20°F), low-temperature bolting material (such as ASTM A 320) shall be used. |

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| 2.9.2 **Castings**<br><br>2.9.2.1 Castings shall be sound and free from hot tears, shrink holes, blow holes, cracks, scale, blisters, and similar injurious defects. Porosity shall not exceed the limits stated in the material inspection acceptance criteria (4.2.2).<br><br>Surfaces of castings shall be cleaned by sandblasting, shot blasting, chemical cleaning, or any other standard method. Mold-parting fins and remains of gates and risers shall be chipped, filed, or ground flush.<br><br>2.9.2.2 The use of chaplets in pressure castings shall be held to a minimum. The chaplets shall be clean and corrosion free (plating permitted) and of a composition compatible with the casting.<br><br>2.9.2.3 Ferrous castings for pressure retaining parts shall not be repaired by welding, peening, plugging, burning in, or impregnating, except as specified in 2.9.2.3.1 and 2.9.2.3.2.<br><br>2.9.2.3.1 Weldable grades of steel castings may be repaired by welding, using a qualified welding procedure based on the requirements of Section VIII, Division 1, and Section IX of the ASME Code.<br><br>2.9.2.3.2 Cast gray iron or nodular iron may be repaired by plugging within the limits specified in ASTM A 278, A 395, or A 536. The holes drilled for plugs shall be carefully examined, using liquid penetrant, to ensure that all defective material has been removed. All repairs that are not covered by ASTM specifications shall be subject to the purchaser's approval.<br><br>2.9.2.4 Fully enclosed cored voids, including voids closed by plugging, are prohibited.<br><br>2.9.2.5 Grey cast iron (ASTM A278) shall not be used for pressure containing parts that handle flammable or toxic fluids. With the purchaser's approval, nodular cast iron (ASTM A 395) may be used in such services. | **6.13.3 Castings**<br><br>6.13.3.1 Castings shall be sound and free from porosity, hot tears, shrink holes, blow holes, cracks, scale, blisters, and similar injurious defects in excess of that specified in the material specification or any additional specified acceptance criteria (See 8.2.2).<br><br>6.13.3.2 Surfaces of castings shall be cleaned by sandblasting, shot-blasting, chemical cleaning, or other standard methods to meet the visual requirements of MSS-SP-55. Mold-parting fins and the remains of gates and risers shall be chipped, filed or ground flush.<br><br>6.13.3.3 The use of chaplets in pressure castings shall be held to a minimum. If chaplets are necessary, they shall be clean and corrosion free (plating is permitted) and of a composition compatible with the casting.<br><br>6.13.3.4 Ferrous pressure-containing castings shall not be repaired by welding, peening, plugging, burning in, or impregnating, except as follows:<br><br>a) Weldable grades of steel castings may be repaired by welding in accordance with 6.13.5. Weld repairs shall be inspected according to the same quality standard used to inspect the casting.<br><br>b) All other repairs shall be subject to the purchaser's approval.<br><br>6.13.3.5 Fully enclosed cored voids, which become fully enclosed by methods such as plugging, welding, or assembly, are prohibited. |

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| Note: It is recommended that nodular cast iron only be used for services less than 200 psig (14 bar gauge) and 125°F (50°C).<br><br>2.9.2.6 Nodular iron castings shall be produced in accordance with ASTM A 395. The production of the castings shall also conform to the conditions specified in 2.9.2.6.1 through 2.9.2.6.5.<br><br>2.9.2.6.1 A minimum of one set (three samples) of Charpy V-notch impact specimens at one-third the thickness of the test block shall be made from the material adjacent to the tensile specimen on each keel or Y block. These specimens shall have a minimum impact value of 14 joules (10 foot-pounds) at room temperature.<br><br>2.9.2.6.2 The keel or Y block cast at the end of the pour shall be at least as thick as the thickest section of the main casting.<br><br>2.9.2.6.3 Integrally cast test bosses, preferably at least 25 mm (1 inch) in height and diameter, shall be provided at critical areas of the casting for subsequent removal for the purposes of hardness testing and microscopic examination. Critical areas are typically heavy sections, section changes, high-stress points such as drilled lubrication points, the cylinder bore, valve ports, flanges, and other points agreed upon by the purchaser and the vendor. Classification of graphite nodules shall be in accordance with ASTM A 247.<br><br>2.9.2.6.4 An as-cast sample from each ladle shall be chemically analyzed.<br><br>2.9.2.6.5 Brinell hardness readings shall be made on the actual casting at feasible locations on section changes, flanges, the cylinder bore, and valve ports. Sufficient surface material shall be removed before hardness readings are made to eliminate any skin effect. Readings shall also be made at the extremities of the casting at locations that represent the sections poured first and last. These shall be made in addition to Brinell readings on the keel or Y blocks. | • 6.13.3.6 If specified, for casting repairs made in the supplier's shop, repair procedures including weld maps shall be submitted for purchaser's approval. The purchaser shall specify if approval is required before proceeding with repair. Repairs made at the foundry level shall be controlled by the casting material specification ("producing specification").<br><br>6.13.3.7 Pressure-retaining castings of carbon steel shall be furnished in the normalized |

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<td>Unless otherwise agreed upon by the purchaser and the vendor, the forging material shall be selected from those listed in Appendix B.<br><br>2.9.4 Welding<br><br>2.9.4.1 Welding of piping and pressure-containing parts, as well as any dissimilar metal welds and weld repairs, shall be performed and inspected by operators and procedures qualified in accordance with Section VIII, Division 1, and Section IX of the ASME Code. The manufacturers data report forms as specified in the code are not required.</td>
<td>and tempered condition.<br><br>6.13.4 Forgings<br><br>6.13.4.1 Pressure-containing ferrous forgings shall not be repaired except as follows:<br><br>a) Weldable grade of steel forgings may be repaired by welding in accordance with 6.13.5. After major weld repairs, and before hydro test, the complete forging shall be given a post-weld heat treatment to ensure stress relief and continuity of mechanical properties of both weld and parent metal.<br><br>b) All repairs that are not covered by the material specification shall be subject to the purchaser’s approval<br><br>6.13.5 Welding<br><br>• 6.13.5.1 Welding and weld repairs shall be performed in accordance with Table 3. If specified, alternative standards may be proposed by the supplier for the purchaser’s approval and, if so, they shall be referenced in the data sheets (see Annex A).<br><br>Table 3 — Welding requirements<br><br>
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<tr><td>Requirement</td><td>Applicable Code or Standard</td></tr>
<tr><td>Welder/operator qualification</td><td>ASME BPVC Section IX or EN287</td></tr>
<tr><td>Welding procedure qualification</td><td>Applicable material specification or, where weld procedures are not covered by the material specification ASME BPVC Section IX or EN287 or EN 288</td></tr>
<tr><td>Non-pressure retaining<br>Structural welding such as mounting plates or supports</td><td>AWS D1.1</td></tr>
<tr><td>Magnetic particle or liquid penetrant examination of the plate edges</td><td>ASME BPVC Section VIII, Division 1, UG-93(d)(34)</td></tr>
<tr><td>Post-weld heat treatment</td><td>Applicable material specification or ASME BPVC Section VIII, Division 1, UW 40</td></tr>
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| | 6.13.5.2 The vendor shall be responsible for the review of all repairs and repair welds to ensure that they are properly heat treated and nondestructively examined for soundness and compliance with the applicable qualified procedures (see 6.13.5.1). Repair welds shall be nondestructively tested by the same method used to detect the original flaw; however, the minimum level of inspection after the repair shall be by the magnetic particle method in accordance with 8.2.2.1. |
| 2.9.4.2 The vendor shall be responsible for the review of all repairs and repair welds to ensure that they are properly heat treated and nondestructively examined for soundness and compliance with the applicable qualified procedures (2.9.1.11). Repair welds shall be nondestructively tested by the same method used to detect the original flaw. As a minimum, the inspection shall be in accordance with 4.2.2.4 or 4.2.2.5 as applicable. | |
| 2.9.4.3 Unless otherwise specified, all welding other than that covered by Section VIII, Division 1, of the ASME Code and ASME B31.3, such as welding on baseplates, standard pump items, non pressure ducting, lagging, and control panels, shall be performed in accordance with AWS D1.1. | 6.13.5.3 Pressure-containing parts made of wrought materials or combinations of wrought and cast materials shall conform to the conditions specified in 6.13.5.3 a) through and 6.13.5.3. c) . These requirements do not apply to casing nozzles and auxiliary connections; see 6.13.5.4. |
| 2.9.4.4 Unless otherwise specified, pressure-containing components made of wrought materials or combinations of wrought and cast materials shall conform to the conditions specified in 2.9.4.4.1 through 2.9.4.4.4. | a) Accessible surfaces of welds shall be inspected by magnetic particle or liquid penetrant examination after back chipping or gouging and again after post-weld heat treatment or, for austenitic stainless steels, after solution annealing. If specified, the quality control of welds that will be inaccessible on completion of the fabrication shall be agreed on by the purchaser and vendor prior to fabrication. |
| 2.9.4.4.1 Plate edges shall be inspected by magnetic particle or liquid penetrant examination as required by Section VIII, Division 1, UG-93(d)(3), of the ASME Code. | b) Pressure-containing welds, including welds of the casings to axial-joint and radial-joint flanges, shall be full-penetration welds. |
| 2.9.4.4.2 Accessible surfaces of welds shall be inspected by magnetic particle or liquid penetrant examination after back-chipping or gouging and again after post weld heat treatment. | |
| 2.9.4.4.3 Pressure-containing welds, including welds of the cylinder to horizontal- and vertical- joint flanges, shall be full-penetration welds. | c) Where dimensional stability of the component must be assured for the integrity of pump operation, post-weld heat treatment shall be performed regardless of thickness. |
| 2.9.4.4.4 Cylinders fabricated from materials that, according to Section VIII, Division 1, of | 6.13.5.4 Connections welded to casings shall be installed as follows: |

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| the ASME Code, require post-weld heat treatment, shall be heat treated regardless of thickness.<br><br>2.9.4.5 Connections welded to pressure casings shall be installed as specified in 2.9.4.5.1 through 2.9.4.5.5.<br><br>2.9.4.5.1 In addition to the requirements of 2.9.4.1, the purchaser may specify that 100 percent radiography, magnetic particle inspection, or liquid penetrant inspection of welds is required.<br><br>2.9.4.5.2 Auxiliary piping welded to chromium-molybdenum alloy steel or 12 percent chrome steel components shall be of the same material, except that chromium-molybdenum alloy steel pipe may be substituted for 12 percent chrome steel pipe.<br><br>2.9.4.5.3 When heat treating is required, piping welds shall be made before the component is heat treated.<br><br>2.9.4.5.4 When specified, proposed connection designs shall be submitted to the purchaser for approval before fabrication. The drawing shall show weld designs, size, materials, and pre-weld and post weld heat treatments.<br><br>2.9.4.5.5 All welds shall be heat treated in accordance with Section VIII, Division 1, UW-40, of the ASME Code.<br><br>**2.9.5 Impact Test Requirements**<br><br>2.9.5.1 To avoid brittle fracture during operation, maintenance, transportation, erection and testing, good design practice shall be followed in the selection of fabrication methods, welding procedures and materials for vendor furnished steel pressure retaining parts that may be subject to a temperature below the ductile-brittle transition point. | a) Attachment of suction and discharge nozzles shall be by means of full fusion, full-penetration welds. Weld neck flanges, or socket weld flanges if approved by the purchaser, shall be used for pumps handling flammable or hazardous liquids. Welding of dissimilar metals shall not be performed.<br><br>b) If specified, proposed connection designs shall be submitted for purchaser approval before fabrication. The drawings shall show weld designs, size, materials, and pre and post-weld heat treatments.<br><br>c) Post-weld heat treatment, if required, shall be carried out after all welds, including piping welds, have been completed.<br><br>d) Unless otherwise specified, auxiliary piping welded to alloy steel casings shall be of a material with the same nominal properties as the casing material.<br><br>**6.13.6 Low Temperature Service**<br><br>• 6.13.6.1 The purchaser shall specify the minimum design metal temperature and concurrent pressure that the pump will be subjected to in service. This temperature shall be used to establish impact test requirements.<br><br>Note: Normally, this will be the lower of the minimum surrounding ambient temperature or minimum fluid pumping temperature; however, the purchaser may specify a minimum design metal temperature based on properties of the pumped fluid, such as auto-refrigeration at reduced pressures.<br><br>6.13.6.2 To avoid brittle failures, materials and construction for low temperature service |

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| Note: The published design-allowable stresses for many materials in the ASME Code and ANSI standards are based on minimum tensile properties. The ASME Code and ANSI standards do not differentiate between rimmed, semi-killed, fully-killed, hot-rolled and normalized material, nor do they take into account whether materials were produced under fine- or coarse-grain practices. The vendor shall exercise caution in the selection of materials intended for service between -30°C (-20°F) and 40°C (100°F). | shall be suitable for the minimum design metal temperature in accordance with the codes and other requirements specified. The purchaser and the vendor shall agree on any special precautions necessary with regard to conditions that may occur during operation, maintenance, transportation, erection, commissioning and testing.<br><br>6.13.6.3 Care shall be taken in the selection of fabrication methods, welding procedures, and materials for vendor provided steel pressure retaining parts that may be subject to temperatures below the ductile brittle transition temperature. The published design-allowable stresses for materials in internationally recognized standards such as the ASME Code and ANSI standards are based on minimum tensile properties. Some standards do not differentiate between rimmed, semi-killed, fully killed, hot-rolled, and normalized material, nor do they take into account whether materials were produced under fine- or course-grain practices. Therefore the vendor should exercise caution in the selection of materials intended for services between –29°C (–20°F) and 40°C (100°F).<br><br>• 6.13.6.4 The purchaser shall specify whether ASME BPVC Section VIII, Division 1, shall apply with regard to impact-testing requirements.<br><br>• 6.13.6.5 If ASME BPVC Section VIII, Division 1, is specified (see 6.14.6.3), the following shall apply:<br><br>a) all pressure-retaining steels applied at a specified minimum design metal temperature below –29 °C (–20 °F) shall have a Charpy V-notch impact test of the base metal and the weld joint unless they are exempt in accordance with ASME BPVC Section VIII, Division 1, UHA-51;<br><br>b) carbon steel and low alloy steel pressure-retaining parts applied at a specified minimum design metal temperature between –29°C (– 20°F) and 40°C (100°F) shall require impact testing as stated below<br><br>1) Impact testing is not required for parts with a governing thickness of 25 mm (1 in) or less.<br><br>2) Impact testing exemptions for parts with a governing thickness greater than 25 mm (1 in) shall be established in accordance with paragraph UCS-66 in ASME BPVC Section VIII, Division 1. Minimum design metal temperature without impact |

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| | testing may be reduced as shown in Figure UCS-66.1. If the material is not exempt, Charpy V-notch impact test results shall meet the minimum impact energy requirements of paragraph UG-84 of the ASME Code.<br><br>6.13.6.6 Governing thickness used to determine impact testing requirements shall be the greater of the following:<br><br>a) the nominal thickness of the largest butt welded joint;<br><br>b) the largest nominal section for pressure containment, excluding:<br>1) structural support sections such as feet or lugs;<br>2) sections with increased thickness required for rigidity to mitigate deflection;<br>3) structural sections required for attachment or inclusion of mechanical features such as jackets or seal chambers<br><br>c) one fourth of the nominal flange thickness, (in recognition that the predominant flange stress is not a membrane stress). |
| 2.9.5.2 All pressure containing components including nozzles, flanges and weldments shall be impact tested in accordance with the requirements of Section VIII, Division 1, sections UCC-65 through 68, of the ASME Code. High-alloy steels shall be tested in accordance with Section VIII, Division 1, Section UHA-51, of the ASME Code. Impact testing is not required if the requirements of Section VIII, Division 1, Section UG-20F, of the ASME Code are met. Nominal thickness for castings as defined in Section VIII, Division 1, Paragraph UCS-66(2), of the ASME Code shall exclude | |
| 2.10 Nameplates And Rotation Arrows | 6.15 Nameplates And Rotation Arrows |
| 2.10.1 A nameplate shall be securely attached at a readily visible location on the equipment and on any other major piece of auxiliary equipment. | 6.15.1 A nameplate shall be securely attached at a readily visible location on the equipment and on any other major piece of auxiliary equipment. |
| 2.10.2 Rotation arrows shall be cast in or attached to each major item of rotating equipment in a readily visible location. Nameplates and rotation arrows (if attached) shall be of ANSI Standard Type 300 stainless steel or of nickel-copper alloy (Monel or its equivalent). Attachment pins shall be of the same material. Welding is not permitted | 6.15.2 Rotation arrows shall be cast in or attached to each major item of rotating equipment in a readily visible location. Nameplates and rotation arrows (if attached) shall be of ANSI Standard Type 300 stainless steel or of nickel-copper alloy (Monel ®[16] or its equivalent). Attachment pins shall be of the same material. Welding is not permitted. |
| 2.10.3 The purchaser's item number, the vendor's name, the machine's serial number, | 6.15.3 The purchaser's item number, the vendor's name, the machine's serial number, |

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| and the machine's size and type, as well as its minimum and maximum allowable design limits and rating data (including pressures, temperatures, speeds, and power), maximum allowable working pressures and temperatures, hydrostatic test pressures and critical speeds, shall appear on the machine's nameplate. The purchaser will specify on the data sheet whether customary or SI units are to be shown.<br><br>2.11 Quality<br>Refer to API Recommended Practice 683 for guidelines on improving the quality of equipment. | and the machine's size and type, as well as its minimum and maximum allowable design limits and rating data (including pressures, temperatures, speeds, and power), maximum allowable working pressures and temperatures, hydrostatic test pressures and critical speeds, shall appear on the machine's nameplate. The purchaser will specify on the data sheet whether customary or SI units are to be shown. |